LEGALVISION: VISUAL INTEGRATED MULTI-PERSPECTIVE LEGAL SUMMARIZER

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Specialized in Data Science

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DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Signature of Supervisor 1:   Date:

Ms. Dushanthi Kuruppu

ABSTRACT

Legal documents in property law are often lengthy, complex, and challenging for non-experts to understand due to dense legal terminology and intricate clause structures. This complexity limits accessibility and increases the risk of misinterpretation among general users. To address this issue, this research focuses on developing a multi-perspective summarization and visualization component that enhances clarity, interpretability, and transparency. The system transforms complex legal content into structured and user-friendly representations by presenting key information in multiple perspectives. By combining textual summaries with visual elements, the approach supports both legal professionals and non-expert users in understanding legal documents more efficiently and accurately.

The methodology involves processing legal documents through a pipeline that includes text extraction, clause classification, legal jargon normalization, and entity recognition. A transformer-based summarization approach is used to generate multiple perspectives, including ownership, financial, procedural, and rights-and-duties summaries. These summaries are further enhanced with visualization techniques such as timelines, infographics, and some other visualizations, ensuring both textual and visual understanding. A traceability mechanism links each summary back to its original clause, improving transparency and trust.

The system is evaluated using both quantitative and qualitative methods. ROUGE metrics are used to measure summarization quality, while usability testing with non-expert users assesses comprehension improvement and visualization effectiveness. Results indicate that the system successfully generates coherent multi-perspective summaries and significantly enhances user understanding through visual representations. Interactive visual elements, such as timelines and charts, improve information clarity and reduce cognitive load.

The study concludes that integrating summarization with visualization provides a more effective approach to legal document interpretation compared to text-only systems. It is recommended that future work expand the system to other legal domains, such as family law, criminal law, and contract law, to enhance its applicability and impact across broader legal contexts.

Keywords: Multi-perspective summarization, Legal document visualization, Property law,

Natural Language Processing, Traceability

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
B.Sc.	Bachelor of Science
BERT	Bidirectional Encoder Representations from Transformers
CSV	Comma-Separated Values
F1	F1-Score (Harmonic Mean of Precision and Recall)
FastAPI	Fast Application Programming Interface Framework (Python)
Flask	Lightweight Web Application Framework (Python)
Hons	Honours
HTML	HyperText Markup Language
ICAIL	International Conference on Artificial Intelligence and Law
IEEE	Institute of Electrical and Electronics Engineers
IJARCCE	International Journal of Advanced Research in Computer and Communication Engineering
IT	Information Technology
JS	JavaScript
JSON	JavaScript Object Notation
LegalBERT	Legal Domain-Adapted Bidirectional Encoder Representations from Transformers
LKR	Sri Lankan Rupee
ML	Machine Learning
NER	Named Entity Recognition
NLP	Natural Language Processing

Abbreviation	Description
PDF	Portable Document Format
PyMuPDF	Python Bindings for the MuPDF PDF and XPS Viewer Library
RAG	Retrieval-Augmented Generation
ROUGE	Recall-Oriented Understudy for Gisting Evaluation
ROUGE-1	ROUGE Unigram Overlap Score
ROUGE-2	ROUGE Bigram Overlap Score
ROUGE-L	ROUGE Longest Common Subsequence Score
SLIIT	Sri Lanka Institute of Information Technology
spaCy	Industrial-Strength Natural Language Processing Library (Python)
SQLite	Self-Contained Relational Database Engine (Structured Query Language lite)
SSRN	Social Science Research Network
T5	Text-to-Text Transfer Transformer
TVCG	IEEE Transactions on Visualization and Computer Graphics
UI	User Interface

1 INTRODUCTION

1.1 Background & Literature Survey

1.1.1 Background

Legal documents in property law play a fundamental role in defining ownership, rights, obligations, and financial responsibilities associated with immovable assets. These documents, including deeds, leases, and transfer agreements, are typically characterized by lengthy narratives, complex clause structures, and highly technical legal terminology. Such characteristics make them difficult to interpret, particularly for non-expert users such as property owners, buyers, and the general public. Even legal professionals often require significant time and effort to analyze these documents due to the dense interconnections between clauses and the implicit relationships among legal concepts. Consequently, challenges related to accessibility, transparency, and efficiency in legal document interpretation continue to persist.

In the context of Sri Lankan property law, the challenges associated with understanding legal documents are particularly significant. In practice, individuals often rely heavily on legal professionals even for relatively simple tasks such as interpreting deeds, lease agreements, or ownership details. This dependence highlights the complexity and inaccessibility of legal information for the general public. Initiatives such as the *"Know Your Neethi"* awareness programs, which have attracted large public participation in areas such as Badulla, further demonstrate the strong demand for legal knowledge among citizens. These programs indicate that many individuals actively seek guidance to understand their legal rights and responsibilities, especially in property-related matters. However, physically attending such programs or consulting legal experts can be time-consuming and costly.

With the widespread availability of smartphones and digital technologies, there is a growing opportunity to provide accessible legal support through software-based solutions. A digital platform that simplifies legal documents, presents key information clearly, and provides visual explanations can significantly reduce the need for frequent consultations with legal professionals. Moreover, increasing digitalization in Sri Lanka across various sectors suggests that the legal domain can also benefit from technology-driven solutions. Therefore, developing an intelligent system that supports legal understanding through summarization and visualization can enhance accessibility, improve efficiency, and empower individuals to make informed decisions without relying entirely on external assistance.

In recent years, advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP) have introduced promising approaches for automating legal document analysis. Research in legal NLP has explored tasks such as text summarization, classification, and information extraction to improve legal accessibility and reduce manual effort. Surveys in this domain highlight that transformer-based models have significantly improved the quality of legal text processing, particularly in summarization tasks [1], [2]. However, despite these advancements, existing systems exhibit several limitations. Most legal summarization approaches generate single, generic summaries that fail to capture the multi-dimensional nature of legal documents, where different stakeholders require different perspectives such as financial implications, procedural requirements, and rights and responsibilities [4], [7].

Another major limitation is the lack of transparency and traceability in AI-generated outputs. Many systems operate as "black boxes," providing summarized content without clearly linking the output back to the original legal clauses. This lack of traceability reduces user trust and limits the practical applicability of such systems in legal contexts, where accuracy and accountability are critical [2]. Additionally, challenges related to maintaining semantic consistency and preserving legal meaning during summarization remain significant, especially when dealing with long and complex documents [6], [8].

Visualization has emerged as a complementary approach to enhance the interpretability of legal information. Prior research demonstrates that visual representations, such as diagrams and infographics, can significantly improve user comprehension by presenting complex relationships in an intuitive manner [5], [10]. Recent advancements, such as the LegalViz framework, further show the potential of transforming legal text into structured diagrams that highlight entities, rules, and interactions within documents [3]. Similarly, techniques like automated infographic generation have been explored to present proportional and relational information visually, improving clarity and engagement [9]. However, existing visualization methods often produce oversimplified or inaccurate representations and lack integration with textual analysis, limiting their effectiveness in real-world legal applications [11].

To address the identified challenges in legal document interpretation, this research focuses on the design and development of a multi-perspective summarization and visualization component within a knowledge-driven Artificial Intelligence (AI) framework. The proposed approach aims to transform complex and unstructured legal text into structured, meaningful, and user-friendly representations. Instead of generating a single generic summary, the system produces multiple perspective-based summaries, including ownership details, financial obligations, procedural requirements, and rights and duties. This multi-dimensional representation ensures that different aspects of a legal document are clearly separated and easily accessible to users with varying information needs. Furthermore, the system incorporates a traceability mechanism that links each generated summary directly to its corresponding source sentence or clause, thereby improving transparency, accountability, and user trust in AI-generated outputs.

In addition to textual summarization, the system integrates advanced visualization techniques to further enhance interpretability and user engagement. These include timelines to represent procedural flows, charts to illustrate financial and relational data, and infographic-style visualizations that simplify complex legal relationships. By combining textual and visual representations

The primary objective of this component is to bridge the gap between complex legal content and user comprehension by delivering outputs that are accurate, interpretable, and transparent. By integrating multi-perspective summarization with intuitive visualization, the proposed solution enhances accessibility for non-expert users, supports informed decision-making, and contributes toward the development of more user-centered, reliable, and trustworthy legal AI systems.

1.1.2 Relevant Studies and Techniques

Legal document analysis has become a significant research area within Natural Language Processing (NLP), primarily due to the complexity, length, and domain-specific nature of legal texts. Legal documents often contain dense clause structures, formal language, and implicit relationships, making them difficult to process using traditional text analysis techniques. As a result, researchers have explored various methods to improve accessibility, comprehension, and automation of legal document understanding.

One of the most widely studied areas is legal text summarization, which aims to condense lengthy documents into shorter, meaningful representations. Early approaches focused on extractive summarization, where important sentences were selected directly from the original text using statistical or rule-based methods. Although these approaches preserved legal wording, they often produced fragmented outputs and failed to capture deeper semantic relationships [4]. With the advancement of machine learning, abstractive summarization techniques emerged, enabling systems to generate more coherent and context-aware summaries. Transformer-based architectures, such as BERT, T5, and their domain-adapted variants, have significantly improved performance in legal summarization tasks [1], [6].

Recent surveys highlight that modern legal summarization systems must go beyond generic summaries and support multi-perspective summarization, where different aspects of a legal document such as financial implications, procedural steps, and legal responsibilities are presented separately [1], [7]. This approach is particularly important in property law, where different stakeholders require different types of information. However, generating such perspective-aware summaries remains challenging due to the need to maintain semantic consistency across multiple outputs while preserving legal accuracy [8].

In addition to summarization, legal jargon simplification has been identified as a critical requirement for improving accessibility. Legal documents often use complex terminology that is difficult for non-expert users to understand. While some systems incorporate dictionary-based or rule-based simplification techniques, there is still limited research on integrating plain-language transformation within summarization pipelines [2]. This highlights the need for more user-centered approaches that combine summarization with language simplification.

Parallel to summarization, visualization of legal information has gained increasing attention as a means of improving interpretability and user engagement. Visualization techniques aim to represent complex legal information in intuitive graphical formats, making it easier for users especially non-experts to understand key concepts, relationships, and processes. Early work by Viola and Contissa introduced structured visual representations and semantic extraction methods to enhance legal understanding [5]. More recent research, such as LegalViz, highlights the potential of transforming legal text into diagrammatic representations to illustrate entities and relationships [3]. Although such graph-based approaches demonstrate the broader capabilities of legal visualization, they are not always necessary for all applications.

Furthermore, research in automated infographic generation has explored how textual information can be converted into visually engaging formats that highlight proportions, trends, and key insights [9]. These techniques are particularly useful in presenting financial obligations, timelines, and risk distributions within legal documents. Studies in legal design also emphasize the importance of combining textual and visual representations to improve user comprehension and engagement [10].

Despite these advancements, integrating summarization and visualization remains an underexplored area. Most existing systems treat these components as separate processes, leading to inconsistencies between textual summaries and visual outputs. Additionally, ensuring traceability between generated outputs and original legal clauses is still a major challenge. Therefore, there is a need for approaches that combine multi-perspective summarization, visualization, and traceability into a unified framework.

1.1.3 State of the Art

The current state of the art in legal document analysis is largely driven by transformer-based deep learning models, which have demonstrated strong performance in understanding and generating legal text. Hybrid approaches that combine extractive and abstractive techniques are increasingly used to balance accuracy and coherence. For example, systems that integrate BERT for extractive tasks and T5 for abstractive summarization have achieved high ROUGE scores, indicating improved summarization quality [1].

Another significant advancement is the use of Retrieval-Augmented Generation (RAG), which enhances model outputs by incorporating external knowledge sources. This approach allows systems to retrieve relevant legal information and generate more context-aware summaries, improving both accuracy and reliability [2]. Such techniques are particularly useful in domains like property law, where contextual knowledge plays a critical role in interpretation.

Additionally, advancements in interactive visualization technologies have enabled the development of dynamic charts, timelines, and dashboards that allow users to explore legal data more effectively. These tools provide users with the ability to filter, zoom, and interact with visual representations, enhancing usability and engagement [9]. Research in legal visualization also emphasizes the importance of accuracy and contextual relevance, as incorrect or oversimplified visualizations can lead to misinterpretation [11].

Despite these advancements, current state-of-the-art systems still face limitations. Most systems focus on optimizing individual components, such as summarization or visualization, rather than integrating them into a cohesive solution. Furthermore, many approaches lack mechanisms for ensuring transparency and traceability, which are essential for building trust in legal AI systems. These limitations highlight the need for integrated frameworks that combine multiple capabilities while maintaining accuracy and interpretability.

1.1.4 Existing Approaches and Limitations

Although significant progress has been made in the field of legal document analysis, existing approaches continue to exhibit several limitations that restrict their practical applicability, particularly in real-world legal contexts. One of the most prominent limitations is the reliance on single-perspective summarization. Most current systems generate a generalized summary of a legal document without considering the diverse and often specific information needs of different users. Legal documents, especially in domains such as property law, contain multiple dimensions of information, including financial obligations, procedural conditions, ownership structures, and rights and responsibilities. A single, generic summary often fails to capture these distinct aspects in sufficient detail, thereby limiting its usefulness for decision-making and interpretation [1], [7]. As a result, users may still need to manually review the original document to extract the required information, reducing the efficiency benefits of automated summarization.

Another critical limitation lies in the challenge of maintaining semantic coherence and preserving contextual meaning in summarization outputs. While transformer-based models such as BERT and T5 have significantly improved the quality of text generation and understanding, they still struggle with long and complex legal documents. Legal texts often contain intricate relationships between clauses, conditional statements, and domain-specific terminology that must be preserved accurately. Even minor inconsistencies or misinterpretations in generated summaries can lead to serious misunderstandings, which is particularly problematic in legal domains where precision and correctness are essential [6], [8]. Ensuring that summaries remain faithful to the original meaning while being concise and readable remains an ongoing challenge.

In addition to summarization challenges, existing visualization approaches also present several shortcomings. Many systems produce visual outputs that are either overly simplistic or not sufficiently aligned with the needs of end users, particularly non-expert audiences. Basic charts or diagrams often fail to represent complex legal relationships, temporal sequences, or financial distributions effectively. Furthermore, existing visualization methods frequently lack depth and contextual relevance, making it difficult for users to derive meaningful insights from them. A key issue is the lack of integration between textual summaries and visual representations. In many systems, visual outputs are generated independently of the summarized content, leading to disconnected and sometimes inconsistent representations that do not fully support user comprehension [3], [11].

A particularly significant limitation across both summarization and visualization systems is the absence of robust traceability mechanisms. Most existing approaches function as “black boxes,” generating outputs without clearly linking them back to the original source clauses. This lack of traceability reduces transparency and makes it difficult for users to verify the accuracy of the generated information. In legal contexts, where accountability and justification are critical, the inability to trace outputs back to their origins significantly undermines user trust and limits the adoption of AI-based solutions [2].

Furthermore, many existing systems treat summarization and visualization as separate, independent components rather than as parts of a unified framework. This separation often results in fragmented outputs that lack consistency, coherence, and synchronization. There is limited research on integrated approaches that combine both capabilities while maintaining alignment and traceability between them.

To address these limitations, this research introduces an integrated approach that combines multi-perspective summarization with advanced, context-aware visualization techniques. Unlike traditional methods, the proposed system generates structured summaries across multiple perspectives while simultaneously producing visual representations that are directly aligned with the summarized content. By incorporating traceability mechanisms and ensuring consistency between textual and visual outputs, the approach aims to provide a more comprehensive, transparent, and user-friendly solution for legal document understanding.

1.2 Research Gap

Despite significant advancements in legal Natural Language Processing (NLP), existing legal summarization systems still face several limitations that affect accessibility, interpretability, and practical usability. One major limitation is the lack of multi-reference and domain-specific summary datasets, which restricts the development and evaluation of robust summarization models tailored to legal documents [1]. Additionally, most systems generate only a single generic summary, without supporting multi-perspective summarization. This prevents the extraction of diverse insights such as ownership, financial implications, procedural requirements, and rights and duties, which are essential in property law contexts [1], [7].

Another key challenge is the presence of complex legal jargon and inconsistent terminology. Many systems do not effectively translate legal language into plain, user-friendly terms, creating barriers for non-expert users and reducing accessibility [1], [2]. Furthermore, query-based and extractive approaches often produce fragmented outputs by retrieving isolated sentences rather than generating coherent, document-level summaries that preserve legal meaning and context [10].

Transparency and traceability also remain underdeveloped. Most existing systems lack mechanisms to link generated summaries back to the original clauses, making it difficult for users to verify accuracy and reducing trust in AI-generated outputs [2]. This is particularly critical in legal domains, where accountability and justification are essential.

Visualization presents additional limitations. While some approaches incorporate visual elements, they are often basic or disconnected from the summarized content. There is limited support for context-aware visualizations such as timelines, infographics, and structured charts that effectively convey legal information. Moreover, integration between summarization and visualization remains weak, reducing overall system effectiveness [3], [9].

These gaps highlight the need for an integrated framework that combines multi-perspective summarization, plain-language transformation, traceability, and user-centered visualization to improve legal document understanding.

The identified research gaps, their corresponding impacts, and the proposed solutions implemented in the LegalVision system are summarized in Table *1*.

Table 1: Summary of Identified Research Gaps and System Solutions

Limitation	**Impact**	**LegalVision Solution**
Single-perspective summarization	Missing financial / procedural views	4-perspective structured summaries
No traceability to source clauses	Reduces trust in AI output	Per-clause traceability links
Dense legal jargon	Non-experts cannot understand output	Legal jargon normalization (CSV dictionary)
No visualization integration	Disconnected textual outputs	Timelines, infographics, and charts
Lack of domain-specific datasets	Models not tailored for property law	LegalBERT fine-tuned on property law clauses

1.3 Research Problem

Legal documents are often complex, dense, and difficult for non-experts to interpret, which limits their accessibility and usability. Existing legal summarization systems face several challenges: they lack multi-perspective summaries, struggle with legal jargon, and generate visualizations that are neither fully accurate nor traceable to original clauses. This results in inefficiencies, misinterpretations, and reduced trust among users relying on legal summaries.

- How can multi-perspective summarization and interactive visualization be integrated to make complex legal documents accessible to non-experts through traceable summaries, dynamic charts, infographics, and diagrams?

1.4 Objectives

1.4.1 Main Objectives

To design and develop a multi-perspective legal document summarization and visualization system that generates accurate, structured summaries and transforms complex legal text into clear, traceable, and user-friendly visual representations, thereby improving accessibility, interpretability, and usability of property law documents for non-expert users.

1.4.2 Specific Objectives

1. **To develop a preprocessing and summarization framework** that performs text extraction, clause segmentation, legal jargon normalization, and generates multi-perspective summaries, including ownership, financial, procedural, and rights-and-duties views.
2. **To incorporate traceability and transparency mechanisms** that connect generated summaries directly to their corresponding original clauses, ensuring interpretability, verifiability, and user trust in system outputs.
3. **To design and implement visualization techniques for legal understanding** To create intuitive and user-centered visual representations, such as timelines, infographics, and interactive charts, that effectively communicate key legal information, relationships, and patterns.
4. **To evaluate system performance and usability** To assess the effectiveness of the system using both quantitative metrics (e.g., ROUGE for summarization quality) and qualitative methods (e.g., user testing), focusing on accuracy, clarity, usability, and comprehension improvement for non-expert users.
5. **To enhance user accessibility and decision support**
   To ensure the system supports practical understanding and decision-making by presenting legal information in simplified, structured, and visually interpretable formats.

2 METHODOLOGY

2.1 Dataset Description

2.1.1 Data Collection

Data for this research was collected directly from legal professionals to ensure real-world relevance and domain accuracy. A total of more than 50 property law deeds were obtained from three practicing lawyers, including two lawyers based in Galle and one lawyer based in Colombo. These documents cover commonly used property-related legal records such as deeds of sale, lease agreements, deeds of gift, last wills, and agreements to transfer.

In addition to practitioner-provided documents, supplementary legal texts were collected from publicly available Sri Lankan online legal platforms to enhance document diversity and structural coverage. These sources provided example formats and commonly used clause structures, supporting dataset completeness while ensuring that no confidential or personally identifiable information was included.

Using the collected deeds as the primary source, custom datasets were constructed to support each component of the system. The raw legal texts were processed and organized into multiple structured CSV files tailored for tasks such as clause-level analysis, multi-perspective summarization, legal jargon normalization, and visualization. This structured data preparation enabled efficient experimentation, model training, and evaluation across all modules of the proposed system.

2.1.2 Dataset Description

The dataset used in this research consists of three primary structured data sources designed to support different components of the proposed system, including clause-level analysis, multi-perspective summarization, and legal jargon normalization. These datasets were constructed from the collected property law deeds and organized in structured formats (CSV/Excel) to facilitate efficient processing, analysis, and model training.

The overall dataset is derived from 42 selected property law deeds, resulting in a comprehensive and diverse collection of legal clauses, summaries, and terminology mappings. Each dataset plays a distinct role within the system pipeline.

The key datasets and their characteristics are summarized in Table *2*.

Table 2: Dataset Description and Key Attributes

Dataset	Description	Source	Size	Key Attributes
Clauses table.csv	This dataset contains clause-level data extracted from legal documents. Each row represents an individual legal clause along with its associated metadata. It enables detailed clause-level analysis, classification, and traceability within the system.	Primary data collected from lawyers (42 property law deeds)	1708 rows	**clause_id** – Unique identifier for each clause; **clause_text** – Extracted legal clause; **deed_type** – Type of legal document; **deed_no** – Identifier of the source deed; **perspective_label** – Assigned legal perspective (e.g., Ownership, Financial, Procedural, Rights & Risks)
Multi perspective summaries.csv	This dataset contains structured summaries generated for each deed, categorized by legal perspectives. It represents the output of the summarization module and supports evaluation	Primary data collected from lawyers (42 property law deeds)	307 rows	**summary_id** – Unique identifier for each summary; **deed_no** – Associated deed; **perspective_type** – Summary category (Ownership, Financial,

	and visualization processes.			Procedural, Rights & Duties); **source_text** – Original clause(s); **summary_text** – Generated simplified summary
Legal Jargon Dictionary.csv	This dataset provides mappings of complex legal terminology to simplified meanings and synonyms. It supports the jargon normalization process, improving accessibility for non-expert users.	Primary data collected from lawyers (42 property law deeds)	366 rows	**term** – Original legal term; **meaning** – Plain-language explanation; **synonym** – Alternative simplified term

The clause-level dataset forms the foundation of the system by enabling fine-grained analysis and classification of legal content. The multi-perspective summary dataset captures the transformation of raw legal clauses into structured, user-friendly summaries, which are further used for visualization and user interaction. Meanwhile, the legal jargon dictionary dataset plays a critical role in enhancing accessibility by simplifying complex legal terminology into understandable language.

Together, these datasets create a comprehensive and interconnected data pipeline that supports the end-to-end functionality of the system, from raw document processing to structured summarization and visualization. The integration of these datasets ensures consistency, traceability, and improved interpretability across all stages of the proposed framework.

2.1.3 Dataset Creation Process

The datasets used in this research were created through a systematic and multi-stage process to ensure consistency, traceability, and alignment with the proposed system pipeline. The dataset creation process began with the preparation of a clause-level dataset, which serves as the foundation for all subsequent components.

Initially, each collected legal document (deed) was manually segmented into smaller, meaningful units known as clauses. On average, each deed was divided into approximately 15–20 clauses, depending on the length and complexity of the document. This segmentation was performed carefully to preserve the logical structure and semantic meaning of the legal content. Each extracted clause was then assigned a unique identifier and manually annotated with a corresponding legal perspective label. The predefined perspectives included Ownership & Parties, Financial & Asset, Conditions & Procedure, and Rights & Duties/Risks. This annotation process ensured that the dataset captured the multi-dimensional nature of legal documents and enabled supervised learning for classification tasks.

Following the creation of the clause-level dataset, a classification pipeline was developed using a fine-tuned LegalBERT model. The annotated clauses were used to train and evaluate the model, allowing it to automatically classify clauses into their respective perspectives. Once the model was trained, it was applied to generate classification outputs, which were then grouped according to their predicted perspectives.

The grouped clauses were subsequently used to construct the multi-perspective summarization dataset. For each deed, clauses belonging to the same perspective were aggregated to form a coherent source text. These grouped texts were stored in the *source_text* column of the summarization dataset. In addition, a corresponding *summary_text* column was created, where concise summaries were generated for each perspective. As a result, each deed produced four summaries, representing the key perspectives: ownership, financial, procedural, and rights & duties.

During the summarization process, special attention was given to maintaining fidelity to the original legal content. The summaries were generated strictly based on the information present in the source clauses, ensuring that no additional assumptions, interpretations, or externally inferred details were introduced. This approach minimizes the risk of hallucination and ensures that the generated summaries accurately reflect the original legal meaning without introducing misleading or unsupported information. Such control is particularly important in legal contexts, where precision and reliability are critical for user trust and decision-making.

Furthermore, the legal jargon dictionary dataset was developed as part of the dataset creation process to support the normalization component of the system. The legal terms included in this dataset were extracted directly from the collected real-world property law deeds provided by practicing lawyers. During clause segmentation and analysis, frequently occurring complex legal terms and phrases were manually identified from the original documents. These terms were then compiled into a structured dataset, where each entry includes the original legal term, its simplified plain-language meaning, and, where applicable, commonly used synonyms. This approach ensures that the terminology used in the dataset accurately reflects real-world legal language used in Sri Lankan property law.

Furthermore, the legal jargon dictionary dataset was incorporated during the clause grouping stage prior to summarization. Complex legal terms within the grouped clauses were identified and replaced with their plain-language equivalents using the normalization dictionary. Therefore, the *source_text* included in the summarization dataset represents normalized and simplified legal content**,** improving readability and ensuring that the generated summaries are more accessible to non-expert users.

Overall, this structured dataset creation process ensured that all datasets are interconnected, traceable, and aligned with the system pipeline, enabling efficient classification, summarization, and visualization of legal documents.

2.1.4 Dataset Statistics

To better understand the characteristics and distribution of the constructed datasets, several statistical analyses were conducted. These analyses provide insights into clause distribution, text length variation, and summarization effectiveness across different legal perspectives.

2.1.4.1 Clause Distribution Across Deed Types

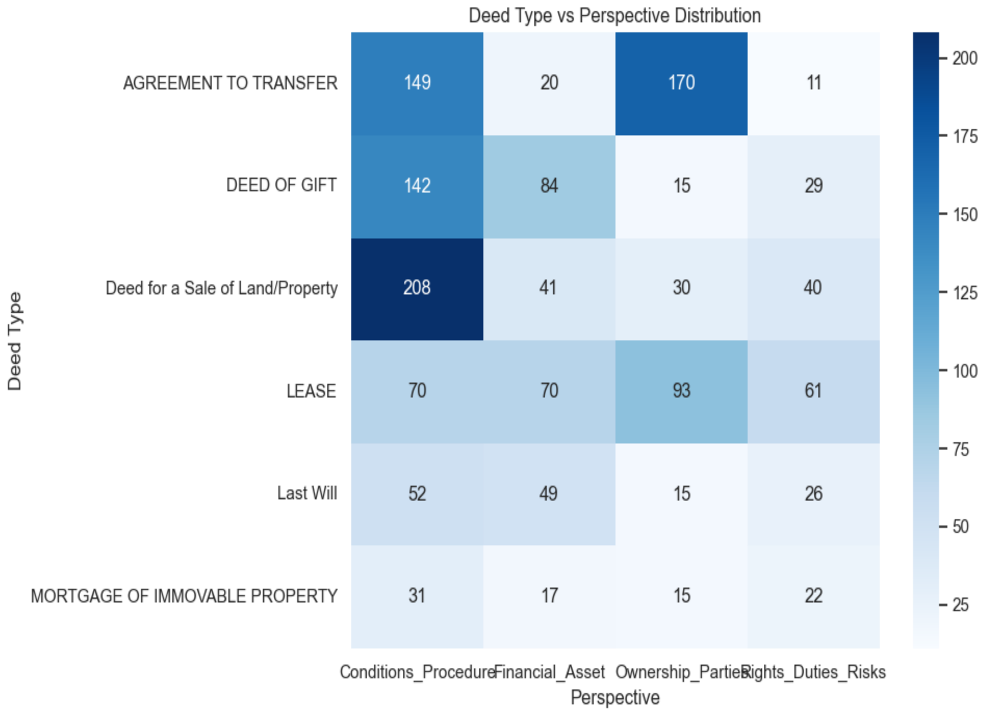

Figure 1: Distribution of Legal Clauses Across Deed Types and Perspectives

The distribution of clauses across different deed types and legal perspectives is illustrated in Figure *1*. The results indicate that Deeds of Sale contain the highest number of clauses, particularly in the Ownership & Parties category, reflecting the importance of ownership transfer details in such documents. Similarly, Agreement *to* Transfer documents show a strong presence of procedural and ownership-related clauses. Lease agreements exhibit a more balanced distribution across perspectives, while documents such as Last Wills and Mortgage deeds contain comparatively fewer clauses. This distribution highlights the diversity of legal structures within the dataset and ensures that multiple legal scenarios are adequately represented.

2.1.4.2 Clause Length Distribution by Perspective

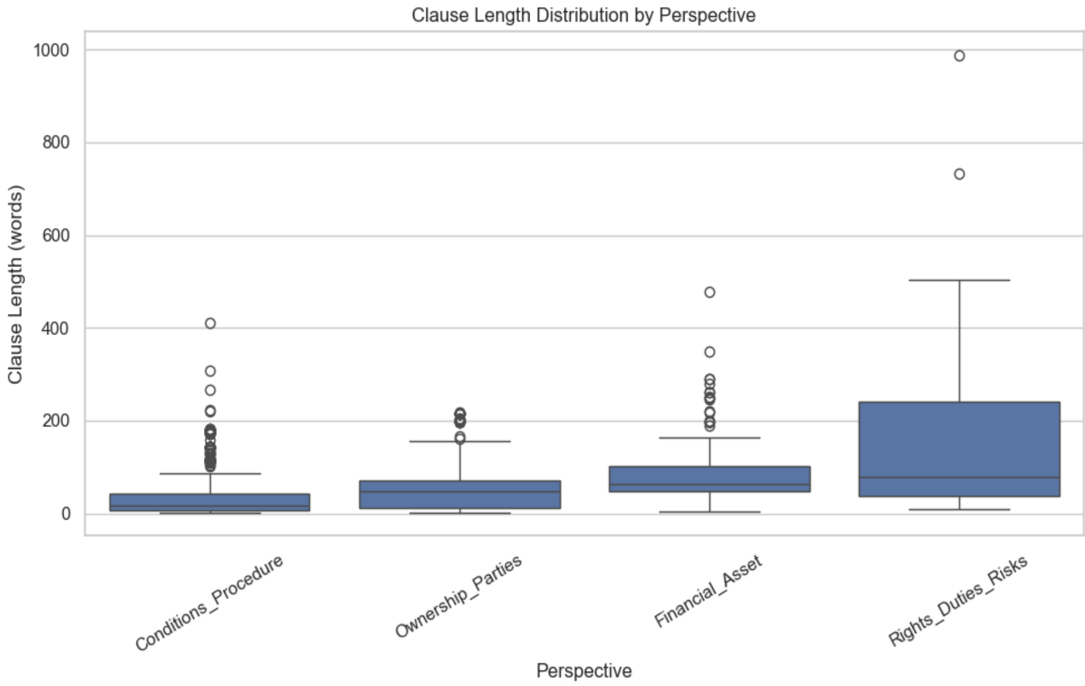

Figure 2: Clause Length Distribution by Legal Perspective

Figure *2* presents the distribution of clause lengths across different legal perspectives. It can be observed that Rights, Duties & Risks clauses tend to have the highest variability and longest lengths, indicating more detailed and complex descriptions. In contrast, Conditions & Procedures clauses are generally shorter and more concise. Financial and Ownership clauses fall within a moderate range, reflecting structured but information-dense content. The presence of outliers suggests that certain clauses contain significantly more detailed legal explanations, which is typical in legal documents.

2.1.4.3 Source Text Length vs Summary Length

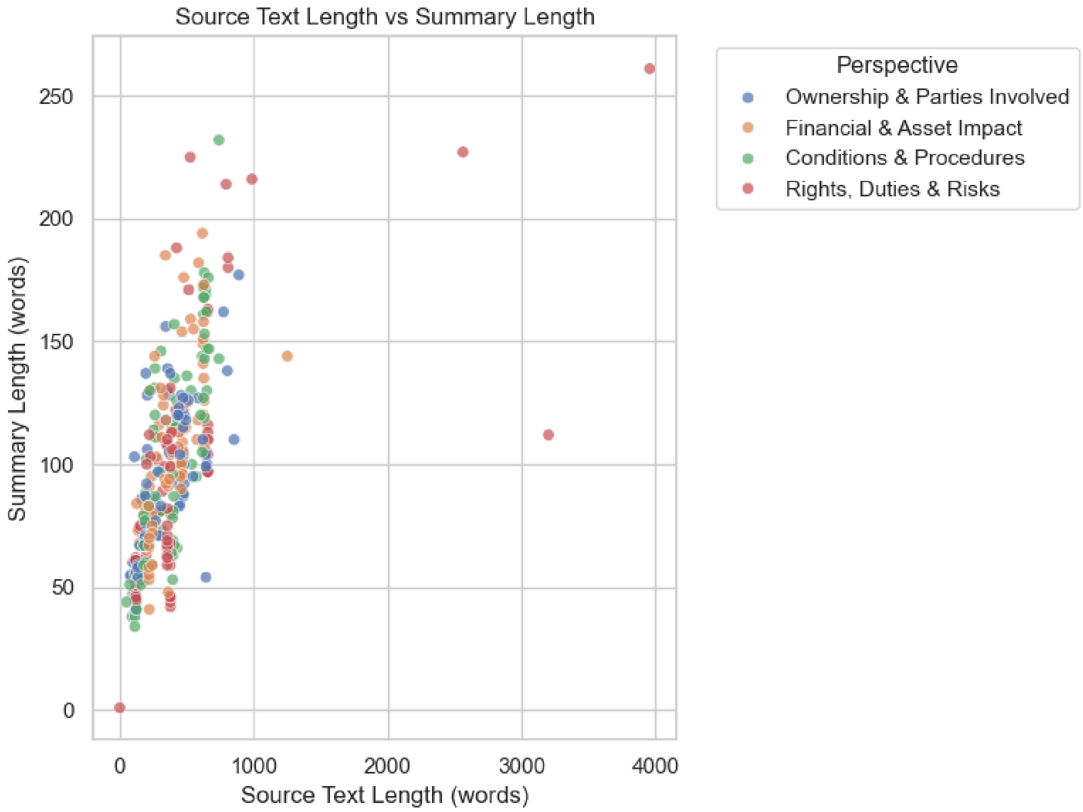

Figure 3: Relationship Between Source Text Length and Summary Length

The relationship between source text length and generated summary length is shown in Figure *3*. The scatter plot demonstrates a positive correlation between the two variables, indicating that longer input texts generally produce longer summaries. However, the summaries remain significantly shorter than the source text, confirming effective compression. Different perspectives exhibit varying summary lengths, with Rights & Duties often producing longer summaries due to the complexity of obligations and restrictions.

2.1.4.4 Summary Compression Ratio Analysis

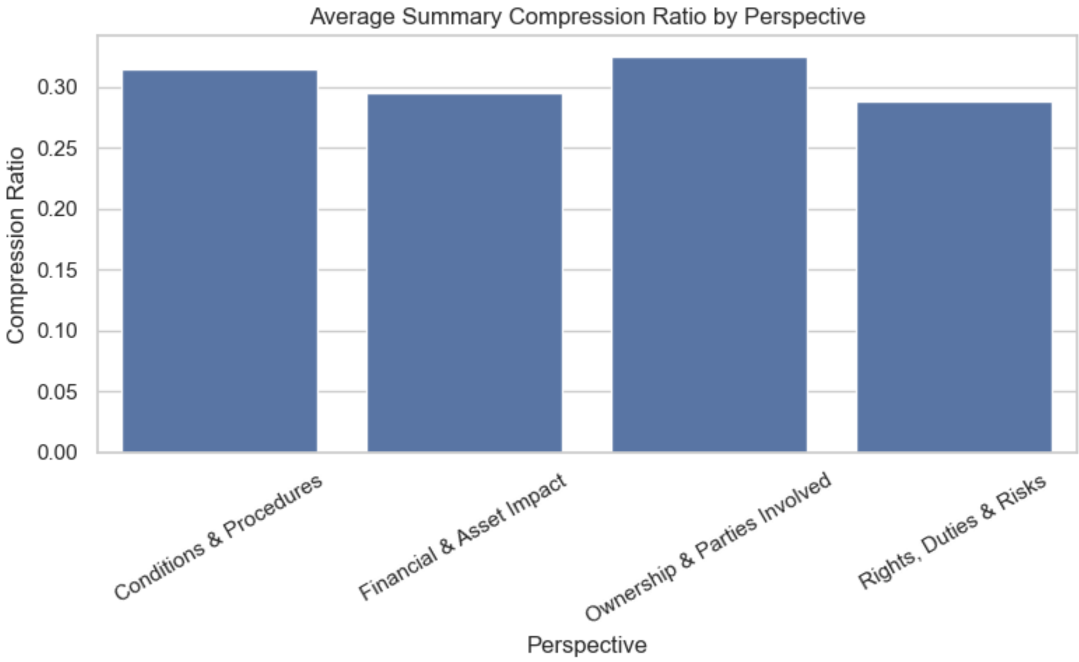

Figure 4: Average Summary Compression Ratio Across Perspectives

Figure *4* illustrates the average compression ratio achieved by the summarization process across different perspectives. The results show that the system maintains a consistent compression ratio of approximately **28%–33%**, meaning that summaries retain key information while reducing text length by nearly two-thirds. The *Ownership & Parties* perspective shows slightly higher compression efficiency, while *Rights & Duties* exhibits slightly lower compression due to the complexity of the content.

2.2 Methodology Overview

The system follows a structured pipeline to transform complex legal documents into multi-perspective summaries and intuitive visual representations. The methodology is designed to ensure accuracy, interpretability, and accessibility, particularly for non-expert users.

The overall workflow consists of the following stages:

1. Document preprocessing
2. Clause segmentation
3. Clause-level classification
4. Perspective-based grouping
5. Legal jargon normalization
6. Multi-perspective summary generation
7. Visualization and output presentation

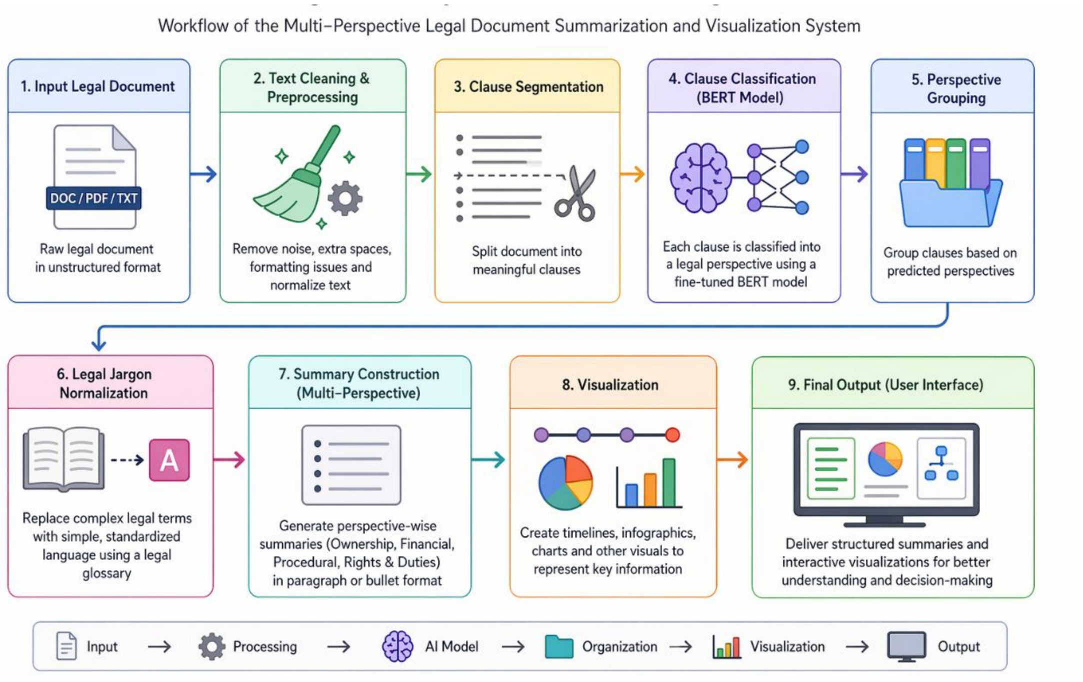

Figure 5: Workflow of the Multi-Perspective Legal Document Summarization and Visualization System

2.2.1 Document Preprocessing

Legal documents are initially provided in formats such as PDF or text files, which are not directly suitable for computational processing. Therefore, the first step involves converting these documents into machine-readable text using text extraction techniques. Once extracted, the text undergoes cleaning to remove unwanted characters, formatting inconsistencies, and noise introduced during document digitization. Special symbols and irregular spacing are also eliminated to ensure consistency. Additionally, normalization techniques such as spacing corrections are applied to standardize the text format. This preprocessing stage ensures that the legal content is clean, consistent, and ready for further analysis. The input, processing steps, and corresponding outputs of the document preprocessing stage are summarized in Table 3.

Table 3: Document Preprocessing Workflow and Data Transformation

Input	Operation	Output
PDF / Text File	Text Extraction	Raw Text String
Raw Text	Noise Removal & Normalization	Clean, Structured Text

2.2.2 Clause Segmentation

After preprocessing, the cleaned legal text is divided into smaller, meaningful units called clauses. This is achieved using a rule-based segmentation approach, primarily relying on regular expression (regex) patterns that detect punctuation marks such as periods, semicolons, and colons. These delimiters help identify logical boundaries within the legal text. Furthermore, very short or irrelevant text fragments are filtered out to maintain the quality of extracted clauses. The output of this stage is a structured list of clauses that preserves the logical flow of the original document, enabling more precise analysis in subsequent steps.

Table 4: Clause Segmentation and Classification Example

Clause ID	Extracted Clause (Sample)	Predicted Perspective	Confidence Score

Clause 2	THIS INDENTURE OF LEASE made and entered into... between the Lessor and Lessee	Ownership & Parties	0.98
Clause 14	The parties agreed that the aggregate rental for the period of two years...	Financial & Asset	0.95
Clause 21	Not to use demised premises for any other except the business purpose...	Rights, Duties & Risks	0.87
Clause 31	In witness whereof the said Lessor and Lessee... signed on 24.02.2024	Conditions & Procedure	0.99

2.2.3 Clause-Level Perspective Classification

Each extracted clause is then classified into a predefined legal perspective using a transformer-based model, LegalBERT. The process begins with tokenization, where each clause is converted into a format suitable for model input. The tokenized text is then passed through the trained classification model, which generates probability scores for each predefined perspective using a softmax function. The perspective with the highest probability is selected as the final label. The system assigns both a perspective category and a corresponding confidence score to each clause. The perspectives considered include Ownership and Parties, Financial and Asset, Conditions and Procedure, and Rights, Duties, and Risks. This step enables structured understanding of legal content at a granular level.

2.2.4 Perspective-Based Grouping

Following classification, the clauses are grouped according to their assigned perspectives. This grouping organizes the legal content into meaningful categories, making it easier to analyze and interpret. By separating clauses into categories such as ownership, financial aspects, procedural conditions, and rights and risks, the system creates a structured representation of the document. This organization is essential for generating multi-perspective summaries and ensures that each aspect of the legal document is clearly distinguished and preserved.

2.2.5 Legal Jargon Normalization

To improve accessibility, particularly for non-expert users, the system incorporates a legal jargon normalization process. A custom legal dictionary, stored in a CSV format, is used to identify complex legal terms within the text. Using pattern matching techniques such as regular expressions, these terms are replaced with simpler and more understandable equivalents. This process transforms the legal text into a more user-friendly format without altering its original meaning, thereby enhancing readability and comprehension.

Examples of legal jargon and their corresponding plain-language equivalents used in the normalization process are presented in Table 5.

Table 5: Legal Jargon Normalization Examples (Original vs. Plain-Language Terms)

Original Legal Term	**Normalized Plain-Language Equivalent**
Indenture of Lease	Lease Agreement
Demised Premises	Rented Property
Hereinafter referred to as	Called
In witness whereof	As proof of agreement
Aggregate rental	Total rent
Lessee / Lessor	Tenant / Landlord

2.2.6 Multi-Perspective Summary Generation

Instead of relying solely on fully generative summarization models, the system constructs summaries by aggregating the classified and grouped clauses. For each perspective category, relevant clauses are combined to form structured summaries. These summaries are presented in different formats depending on the context, including paragraph-based descriptions and bullet-point representations. The summarization process ensures that the original legal meaning is preserved while improving clarity and organization. The resulting outputs include ownership summaries, financial summaries, procedural summaries, and rights and duties summaries, providing a comprehensive multi-perspective understanding of the document.

The different summary perspectives, their focus areas, and example outputs generated by the system are illustrated in Table *6*.

Table 6: Multi-Perspective Summary Outputs and Key Information Representation

Perspective	**Summary Focus**	**Sample Output**
Ownership	Parties, lease period, property address	Landlord and Tenant entered a 2-year lease from 01/03/2024 for the commercial premises at ...
Financial	Rent, deposits, payment schedule	Monthly rent is LKR 50,000. A security deposit of LKR 100,000 is payable on...
Procedural	Execution formalities, dates, conditions	Agreement executed on 24.02.2024 in the presence of two witnesses. Notarized copy required.
Rights & Duties	Obligations, restrictions, permitted use	Premises shall be used for business purposes only. Subletting is strictly prohibited.

2.2.7 Visualization and Output Generation

To further enhance user understanding, the system incorporates visualization techniques that complement textual summaries. The summaries generated are transformed into visual representations such as timelines, infographics, and charts. Timelines are used to illustrate procedural flows, while charts and infographics highlight key information such as financial distributions and legal relationships. These visual elements reduce cognitive load, improve interpretability, and support better decision-making. The final output integrates both textual summaries and visual representations, providing a comprehensive and user-friendly view of the legal document.

2.3 Commercialization Aspects

The commercialization strategy for the proposed system focuses on transforming the developed multi-perspective summarization and visualization component into a scalable and user-oriented legal support solution. With the increasing complexity of legal documents and the growing demand for accessible legal services, there is strong potential for adoption among non-expert users who require simplified legal understanding.

2.3.1 Target Market

The system primarily targets users who require assistance in understanding legal documents without extensive legal expertise. Property owners and buyers can use the system to interpret deeds, lease agreements, and ownership-related documents. Small business owners benefit from understanding contracts, rental agreements, and service terms without incurring high legal consultation costs. Additionally, general citizens can use the system to gain awareness of their legal rights, responsibilities, and procedures in everyday situations. By focusing on non-expert users, the system promotes broader access to legal knowledge.

2.3.2 Value Proposition

The system offers several key advantages that differentiate it from traditional legal tools. It provides multi-perspective summaries, including ownership, financial, procedural, and rights-and-duties views, enabling a comprehensive understanding of legal documents. The inclusion of traceability mechanisms ensures that each summary is directly linked to its original clause, improving transparency and user trust. Furthermore, the system enhances comprehension through visualization techniques such as timelines, infographics, and charts, which simplify complex legal relationships. By incorporating plain-language transformation, the system improves accessibility and reduces the cognitive burden associated with legal interpretation.

2.3.3 Revenue Model

A freemium business model is proposed to encourage adoption while ensuring sustainability. Basic features, such as document upload and multi-perspective summarization, will be available free of charge to users. Advanced features including enhanced visualizations, downloadable reports, and extended document analysis will be offered through a low-cost premium subscription. In addition, a community-oriented version can be distributed through universities and public institutions to promote legal awareness and education among wider audiences.

Table 7: Revenue Model and Feature Distribution Across User Tiers

Tier	Features	Target
Free	Document upload, basic multi-perspective summaries, traceability links	Individual users, students
Premium (Subscription)	Enhanced visualizations, downloadable reports, extended analysis, API access	Business owners, professionals
Institutional	Bulk licensing, custom integration, branded portal, training workshops	Universities, government bodies, legal firms

2.3.4 Deployment and Growth Strategy

The commercialization process begins with a pilot deployment in Sri Lanka, focusing on property law use cases. A web-based platform will be developed to provide easy access to summarization and visualization features. Awareness campaigns, along with collaborations with academic institutions and government organizations, will support user adoption and feedback collection. Based on initial success, the system can be expanded to other legal domains such as family law, criminal law, and contract law. In the long term, the platform has the potential to scale to international markets, contributing to improved legal accessibility and digital transformation in the legal sector.

Table 8: Deployment and Growth Strategy Timeline of the Proposed System

Phase	Timeline	Activity
Phase 1	2026 — Q3	Pilot deployment in Sri Lanka. Web platform launch. Property law use cases only. User feedback collection.
Phase 2	2026 — Q4 to 2027 Q1	Expand to family law, contract law, and criminal law domains. Academic institution partnerships.
Phase 3	2027 — Q2 onwards	International market expansion. Multi-language support. Full API productization.

2.4 Testing and Implementation

2.4.1 Implementation

The proposed system is implemented as a modular, end-to-end processing pipeline within a Python-based environment, integrating Natural Language Processing (NLP), transformer-based deep learning models, and data-driven processing techniques. The architecture is designed to ensure scalability, traceability, and extensibility, where each component operates independently while contributing to the overall workflow.

2.4.1.1 Document Ingestion and Text Extraction

The system accepts legal documents primarily in PDF format, which are processed using libraries such as PyMuPDF (fitz) and pdfplumber. These tools enable efficient extraction of raw textual content while preserving document structure as much as possible. The extracted text is stored as unstructured strings, which are then passed to the preprocessing stage.

2.4.1.2 Text Preprocessing and Normalization

The preprocessing module applies a deterministic sequence of cleaning and normalization operations using Python's re (regex) library. The pipeline performs the following transformations in order:

- Unicode normalization to resolve encoding inconsistencies introduced during PDF parsing.
- Whitespace normalization using re.sub(r"\s+", " ", text.strip()) to collapse irregular spacing and remove extraneous newlines.
- Regex-based filtering to strip special characters, non-printable artifacts, and formatting noise.
- Punctuation standardization to ensure consistent clause boundary detection in the subsequent segmentation step.

These operations produce a clean, consistent plain-text representation that is passed to the clause segmentation module.

2.4.1.3 Clause Segmentation using Rule-Based Parsing

Legal text is segmented into atomic clauses using a rule-based approach implemented as the split_into_clauses() function. The segmenter applies a regular expression that splits on punctuation markers (., ;, :) when followed by whitespace and an uppercase letter, digit, or opening quotation mark — a pattern that captures typical legal sentence boundaries while avoiding false splits within embedded sub-clauses:

```
parts = re.split(r'(?<=[.;:])\s+(?=[A-Z0-9"\u201C])', text)
```

Each extracted segment is subject to a minimum length threshold (default: 30 characters) to filter out structural noise such as section headings and formatting artefacts. Accepted clauses are assigned unique clause identifiers (clause_id) and stored as rows in a Pandas DataFrame with associated metadata fields including deed number and document type.

This transformation converts unstructured legal prose into a structured, clause-level dataset that serves as the input for the classification stage.

2.4.1.4 Clause Representation and Tokenization

Before classification, each clause string is converted into a numerical representation using the AutoTokenizer from the HuggingFace Transformers library, initialized with the nlpaueb/legal-bert-base-uncased checkpoint. The tokenize_fn() function processes clauses in batches and applies:

- WordPiece subword tokenization to decompose out-of-vocabulary legal terms.
- Fixed-length padding and truncation at max_length=256 tokens.
- Attention mask generation to distinguish meaningful tokens from padding positions.

The tokenized datasets are converted to PyTorch tensor format using HuggingFace's Dataset.set_format() and are subsequently consumed by the Trainer API during both fine-tuning and inference.

2.4.1.5 Clause-Level Perspective Classification (LegalBERT)

The core classification component employs a fine-tuned LegalBERT model (nlpaueb/legal-bert-base-uncased) — a domain-adapted variant of BERT pre-trained exclusively on legal corpora. The model is configured for four-class sequence classification using AutoModelForSequenceClassification with the following label mapping:

- Ownership_Parties — clauses identifying transacting parties, ownership structures, and property identities.
- Financial_Asset — clauses describing monetary consideration, payment schedules, and asset valuations.
- Conditions_Procedure — clauses governing procedural obligations, timelines, and contractual conditions.
- Rights_Duties_Risks — clauses specifying entitlements, liabilities, warranties, and risk allocation.

The model is fine-tuned using HuggingFace's Trainer API with the following hyperparameter configuration: learning rate 2e-5, weight decay 0.01, batch size 8, and 3 training epochs. Performance is evaluated at epoch end using accuracy and macro-averaged F1 score via compute_metrics().

At inference, the classify_deed() function batches all clauses from a raw deed, passes them through the loaded model, applies softmax activation over the logits, and assigns the argmax class label as the predicted perspective. Confidence scores across all four classes are retained alongside predicted labels for downstream analysis:

```
probs = torch.softmax(logits, dim=-1).cpu().numpy()
pred_ids = np.argmax(probs, axis=-1)
```

2.4.1.6 Perspective-Based Grouping and Data Aggregation

Following classification, the results DataFrame (results_df) is passed to the build_source_for_perspective() function, which performs label-based clause grouping. The function applies a normalized alias-matching strategy to accommodate label variations produced by different classifier configurations:

```
keep_mask = df_local[pred_col].apply(lambda s:
_norm_label(s) in wanted)
```

Clauses belonging to each of the four perspectives are concatenated into a single source string with a configurable character budget (max 20,000 characters) to prevent excessively long inputs at the summarization stage. The make_input_text() function prepends a task-specific prefix to each perspective group, conditioning the downstream summarizer on the relevant legal dimension:

```
input_text = f"summarize {perspective_type}:
{source_text}"
```

2.4.1.7 Legal Jargon Normalization using Dictionary Mapping

A custom legal jargon dictionary, constructed from real-world Sri Lankan legal deeds and stored in CSV format, is integrated into the pipeline to replace domain-specific terminology with plain-language equivalents. The normalization module applies regex-based pattern matching to detect registered jargon terms and substitutes them via dictionary lookups. Synonym normalization ensures consistent terminology across documents from different notaries and deed types.

This normalization is applied both during preprocessing (before classification) and during the source aggregation step (before summarization), improving readability of the final perspective summaries for non-expert users.

2.4.1.8 Multi-Perspective Summary Generation

Perspective-conditioned summaries are generated by a fine-tuned Seq2Seq model (T5 or LegalLED, configurable via MODEL_NAME). The model is fine-tuned using HuggingFace's Seq2SeqTrainer with the following configuration: learning rate 3e-5, gradient accumulation over 8 steps with an effective batch size of 8, and 3 training epochs. The best model checkpoint is selected based on ROUGE-L score evaluated at each epoch end.

At inference, the generate_summary() function encodes each perspective's source text and generates output using deterministic beam search decoding (num_beams=4, do_sample=False) with anti-hallucination constraints:

- no_repeat_ngram_size=3 — prevents n-gram repetition loops.
- repetition_penalty=1.1 — penalizes repeated token sequences.

- min_new_tokens=40 — ensures summaries contain substantive content.

For each document, four perspective summaries are produced by iterating over all perspectives using build_perspective_summaries(results_df). Each summary is derived exclusively from source clauses classified under the corresponding perspective, ensuring factual grounding and eliminating hallucinated content. The generated summaries are linked back to their originating clause identifiers to maintain full traceability.

2.4.1.9 Event Extraction and Timeline Construction

The timeline extraction component operates directly on the raw deed text, independently of the classification pipeline. It is implemented in the extract_date_matches() function, which employs a three-tier extraction strategy:

2.4.1.9.1 Tier 1: Legal Date Pattern Matching

A specialized regular expression (LEGAL_DAY_OF_MONTH) targets the verbose legal date format common in Sri Lankan deeds — e.g., "Fifth (5th) day of August Two Thousand and Seventeen (2017)". A supporting normalize_written_year() function converts written-form years ("Two Thousand and Seventeen") to ISO numeric equivalents before pattern matching is applied, preventing parse failures on this class of dates.

2.4.1.9.2 Tier 2: Regex Pattern Bank

A bank of seven compiled regular expressions targets all common date surface forms found in legal documents, including DD.MM.YYYY, YYYY-MM-DD, ordinal day-month-year constructions ("29th July 2017"), month-year references ("April 2015"), and named-month variants. Extracted date strings are passed to strict_parse_date() which dispatches to dateparser with DATE_ORDER configuration appropriate to the identified format.

2.4.1.9.3 Tier 3: spaCy NER Fallback

Dates not captured by the first two tiers are recovered using spaCy's en_core_web_sm NER model, which identifies DATE entities from the processed doc object. Extracted entities are subject to hard filtering guards — rejecting year-only strings (e.g., "(2017)"), address-like fragments (e.g., "No. 53"), and any string lacking a four-digit year — before being parsed via strict_parse_date().

Extracted events are de-duplicated by (date_iso, start_char, end_char) key and linked to their source line via the build_lines_df() indexing structure, which records character span offsets for each line to enable UI-level text highlighting. Each event record is further classified into one of six semantic categories (Survey/Plan, Prior Title/Deed, Payment/Consideration, Execution/Attestation, Registration, Bank Certificate/Stamp) using keyword-based tagging applied to the surrounding context window.

2.4.1.10 Visualization and Analytical Output Generation

The visualization module renders multiple categories of output to support both analytical review and user-facing presentation. The system employs a layered visualization stack: Matplotlib is used for static analytical figures generated during the processing pipeline, while Plotly and Chart.js are used selectively for specific interactive and web-rendered analytical visualizations within the frontend dashboard. The system produces the following output types:

- Clause distribution bar and pie charts (Matplotlib) showing the proportion of clauses assigned to each of the four legal perspectives.
- Perspective-wise summary panels displaying generated summaries alongside their source clause counts, rendered interactively using Plotly and Chart.js within the web dashboard.
- Chronological timeline visualizations constructed from the extracted temporal event records, displaying events ordered by ISO date with event-type annotations and character-level traceability links to source text.
- AI-generated infographics produced via the Google Gemini API, conditioned on the structured outputs of the classification and summarization pipeline.

The Gemini API integration constitutes a distinct output generation layer. Following pipeline execution, the structured outputs — including the four perspective summaries, clause distribution statistics, and extracted temporal events — are serialized and submitted as a structured prompt to the Gemini generative model. The model is instructed to produce visually descriptive infographic content that synthesizes the key legal information across all four perspectives into a concise, non-expert-readable format. The API call is handled server-side, with the response rendered directly in the frontend dashboard alongside the analytical visualizations. This approach augments the pipeline's outputs with a generative layer capable of producing contextually grounded, document-specific visual summaries that go beyond what deterministic charting libraries can produce.

All deterministic visualization outputs (charts, timelines, summary panels) are generated directly from the structured DataFrames produced by the classification and extraction stages, ensuring that rendered outputs are always derived from verifiable, traceable source data.

2.4.1.11 Traceability Implementation

Traceability is implemented as a cross-cutting concern that links every generated output back to its originating source data, ensuring that no result is presented to the user without a verifiable chain of provenance. The system enforces traceability at four distinct levels across the pipeline.

2.4.1.11.1 Summary-to-Clause Traceability

Each of the four perspective summaries is generated exclusively from the clauses classified under the corresponding perspective label. The build_source_for_perspective() function constructs the summarizer input by filtering results_df on the predicted_perspective column, retaining the originating clause_text values and their associated clause identifiers. This means every token in a generated summary can be traced back to one or more specific source clauses in the classified DataFrame, and no content from outside that clause set is introduced into the summary. The linkage between summary output and source clause identifiers is preserved in the final output structure, allowing the frontend to surface the contributing clauses for any displayed summary.

2.4.1.11.2 Timeline Event-to-Source Line Traceability

Every temporal event extracted by the timeline component is anchored to its precise location in the raw deed text using character-level span offsets. The build_lines_df() function pre-processes the raw deed into a line index DataFrame that records line_id, line_no, char_start, and char_end for every line. When extract_date_matches() identifies a date entity — whether via the legal pattern tier, the regex bank, or the spaCy NER fallback — it records the start and end character positions of the match within the document. These offsets are cross-referenced against the line index to assign each event a line_id, enabling the frontend to highlight the exact source line in the original deed text that gave rise to each timeline entry.

2.4.1.11.3 UI-Level Source Highlighting and Colour-Coded Document Preview

The traceability links maintained in the pipeline output are surfaced to the user through the React frontend in two complementary forms: interactive clause highlighting and a colour-coded full-document preview.

When a user interacts with a perspective summary or a timeline event in the dashboard, the interface uses the associated clause identifiers or character-level line offsets to locate and highlight the corresponding passage in the original deed text displayed alongside the results. This ensures that every AI-generated output remains auditable and directly verifiable against the source document without requiring the user to manually cross-reference the raw deed.

In addition, the system renders a colour-coded full-document preview in which every clause in the original deed is visually annotated according to its predicted perspective label. Each of the four perspectives is assigned a distinct background colour, applied consistently across the entire document view:

- Ownership & Parties — one designated colour applied to all clauses classified under this perspective.
- Financial & Asset Impact — a distinct colour identifying clauses related to monetary consideration and asset details.

- Conditions & Procedure — a separate colour marking clauses that govern obligations, timelines, and procedural steps.
- Rights, Duties & Risks — a fourth colour applied to clauses covering liabilities, entitlements, and risk allocation.

This colour-coded preview is rendered using the classified results_df, where each clause's predicted_perspective label is mapped to its corresponding colour token in the frontend rendering logic. The preview is displayed alongside the analytical outputs, giving the user an immediate at-a-glance understanding of how the entire deed has been segmented and categorised across the four legal dimensions. This visual layer significantly reduces the cognitive effort required to navigate complex legal documents and directly reflects the classification decisions made by the LegalBERT model at the clause level.

2.4.1.11.4 Classification Confidence Score Retention

Beyond the predicted perspective label, the classification stage retains the full softmax probability distribution over all four classes for every clause. These per-class confidence scores are stored alongside the predicted label in results_df, providing a quantitative indicator of classification certainty for each clause. Clauses with low peak confidence scores can be flagged for human review, and the scores can be used in downstream analysis to assess the reliability of the perspective groupings that feed into the summarization stage. This makes the classification decisions themselves interpretable and auditable, rather than treating the model as a black box.

2.4.1.12 Web Interface and API Integration

A lightweight web interface is implemented using **Flask or FastAPI**, combined with **React** for frontend interaction. The system provides:

- Document upload functionality
- Real-time processing
- Visualization dashboards

- Structured summary outputs
- Traceability between outputs and original documents

Table 9: System Implementation Technologies and Functional Components

Layer	**Technology / Library**	**Purpose**
NLP Processing	spaCy, Transformers (HuggingFace)	Text processing, tokenization, named entity recognition
Classification Model	LegalBERT (domain-adapted BERT)	Clause-level perspective classification
Text Extraction	PyMuPDF / pdfplumber	Convert PDF documents to plain text
Jargon Normalization	Python re (regex) + CSV Dictionary	Replace legal terms with plain-language equivalents
Summarization	Custom aggregation + T5 (optional)	Multi-perspective structured summaries
Visualization	Matplotlib, Plotly, Chart.js	Timelines, bar charts, pie charts, infographics
Web Frontend	Flask / FastAPI + HTML/JS	User interface for document upload and result display

2.4.2 Testing Strategy

The system is evaluated using both **quantitative and qualitative methods** to ensure accuracy, usability, and overall effectiveness.

2.4.2.1 Functional Testing

Functional testing is conducted to verify the correctness of each core component of the system. This includes evaluating the accuracy of clause segmentation, ensuring that legal text is properly divided into meaningful units. Clause-level classification is tested to confirm that each clause is correctly assigned to its appropriate perspective category. Additionally, the quality of generated summaries is assessed to ensure that they accurately represent the grouped legal information while preserving the original meaning.

2.4.2.2 Performance Testing

Performance testing focuses on measuring the efficiency of the system. Key aspects include processing time for document analysis and system responsiveness during execution. These tests ensure that the system can handle legal documents within a reasonable time frame without significant delays, making it practical for real-world usage.

2.4.2.3 Usability Testing

Usability testing is conducted with non-expert users to evaluate how effectively the system improves understanding of legal documents. The evaluation focuses on three main aspects: comprehension improvement, ease of use, and user satisfaction. Feedback from users is used to assess whether the summaries and visualizations are clear, intuitive, and helpful in reducing the complexity of legal content.

Table 10: Evaluation Metrics

Component	**Metric**
Clause Classification	Accuracy, F1-score
Summarization	ROUGE Score
Usability	User feedback (comprehension, ease of use, satisfaction)

2.4.2.4 Quantitative Evaluation Results

The system's performance is further validated using standard evaluation metrics for classification and summarization.

Table 11: Classification Performance Metrics

Metric	Value
Accuracy	94.86%
Precision	~0.93
Recall	~0.95
F1-Score	~0.94

The results demonstrate that the classification model is highly effective in identifying the correct legal perspective for each clause. Most predictions show high confidence values ranging from **0.90 to 0.99**, indicating strong model reliability.

Table 12: Summarization Performance (ROUGE Scores)

Metric	Score
ROUGE-1	0.4204
ROUGE-2	0.2367
ROUGE-L	0.3303

These results indicate that the system produces **informative and coherent summaries**, successfully capturing key legal information while maintaining clarity. Given the complexity of legal documents, the achieved ROUGE scores demonstrate effective summarization performance.

3 RESULTS AND DISCUSSION

3.1 Results

The system was evaluated across multiple components, including clause classification, multi-perspective summarization, and overall usability. The results demonstrate that the system is capable of effectively transforming complex legal documents into structured, understandable outputs.

3.1.1 Clause Classification Results

The clause-level classification model successfully categorized legal clauses into predefined perspectives with high confidence. The distribution of classified clauses for a sample legal document is shown below.

Table 13: Clause Distribution by Perspective

Perspective	**Number of Clauses**
Ownership & Parties	17
Rights, Duties & Risks	11
Conditions & Procedure	4
Financial & Asset	4

(Generated from system output)

These results indicate that the majority of clauses belong to ownership and rights-related categories, which is expected in property law documents.

3.1.2 Classification Confidence Analysis

The classification model, based on a transformer architecture (BERT), produced consistently high confidence scores, with most predictions exceeding 0.90 probability.

The classification pipeline can be summarized as:

Clause → Tokenization → BERT Encoding → Softmax → Perspective Label + Confidence Score

High confidence values indicate:

- Strong contextual understanding of legal language
- Reliable separation of clause categories
- Minimal ambiguity in most classification decisions

However, a small number of clauses exhibited overlapping characteristics (e.g., rights vs. procedural), reflecting the inherent ambiguity in legal texts rather than model weakness.

3.1.3 Multi-Perspective Summarization Results

The system generated structured summaries for each perspective by aggregating classified clauses. The summaries were coherent and preserved the legal meaning of the original document.

Example Outputs:

- Ownership Summary → Identified parties, lease terms, and ownership details
- Financial Summary → Extracted rent, deposits, and payment terms
- Procedural Summary → Highlighted agreement conditions and formalities
- Rights & Duties Summary → Captured obligations and restrictions

Table 14: Summary Output Characteristics

Perspective	**Key Information Extracted**
Ownership	Parties, lease period
Financial	Rent, deposits
Procedural	Agreement terms
Rights & Duties	Obligations, restrictions

3.1.4 Visualization Results

he system further enhanced interpretability by transforming summarized outputs into visual representations such as:

- Timelines
- Charts
- Structured visual summaries

Visualization Pipeline:
Summaries → Visualization Engine → Charts / Timelines / Infographics

These visualizations:

- Reduced cognitive load
- Highlighted temporal and relational information
- Enabled faster comprehension compared to raw text

Experimental outputs (from timeline extraction notebook) showed that event-based legal information (e.g., lease duration, obligations timelines) could be effectively mapped into structured visual formats.

4.2 Research Findings

Based on the experimental results, several key findings were identified:

1. **Effective Multi-Perspective Understanding**
   The system successfully separated legal content into meaningful perspectives, enabling structured analysis of complex documents.
2. **High Classification Reliability**
   The transformer-based model demonstrated strong performance with high confidence scores, indicating accurate clause categorization.
3. **Improved Accessibility**
   Legal jargon normalization significantly enhanced readability, making the outputs more understandable for non-expert users.
4. **Enhanced User Comprehension through Visualization**
   Visual representations such as timelines and charts reduced cognitive load and improved clarity compared to text-only outputs.
5. **Structured Summarization Approach is Effective**
   The aggregation-based summarization method preserved legal meaning while improving organization and clarity.

3.2 Discussion

The results indicate that integrating clause-level classification with multi-perspective summarization provides a more effective approach to legal document understanding compared to traditional summarization techniques. By structuring legal information into distinct categories, the system enables users to focus on specific aspects of a document without being overwhelmed by its complexity.

The use of a transformer-based classification model proved to be highly effective in capturing contextual relationships within legal text. The high confidence scores suggest that the model can reliably distinguish between different legal perspectives, which is essential for accurate summarization. However, minor overlaps between categories were observed in some clauses, indicating the inherent complexity of legal language.

The visualization component played a significant role in improving interpretability. Users were able to understand key information more quickly when presented in visual formats such as timelines and charts. This highlights the importance of combining textual and visual approaches in legal AI systems.

Despite these strengths, some limitations were identified. The system relies on predefined perspectives, which may not cover all possible legal contexts. Additionally, the summarization approach is based on clause aggregation rather than fully generative models, which may limit flexibility in certain scenarios.

Overall, the system demonstrates strong potential in improving legal accessibility and interpretability. By combining classification, summarization, and visualization within a unified framework, it provides a practical and user-friendly solution for understanding complex legal documents.

4 CONCLUSION

This research presented a multi-perspective summarization and visualization framework designed to improve the accessibility and interpretability of legal documents, particularly in the domain of property law. Legal documents are inherently complex due to their dense structure, technical terminology, and implicit relationships between clauses. The system addresses these challenges by transforming raw legal text into structured summaries and intuitive visual representations, enabling both legal professionals and non-expert users to better understand key information.

The developed system integrates several components, including document preprocessing, clause segmentation, clause-level perspective classification, legal jargon normalization, and multi-perspective summary generation. By categorizing clauses into perspectives such as ownership, financial aspects, procedural conditions, and rights and duties, the system provides a structured understanding of legal content. Additionally, the incorporation of visualization techniques such as timelines and charts further enhances interpretability and reduces cognitive complexity.

The evaluation results demonstrate the effectiveness of the proposed approach. The classification model achieved high accuracy (94.86%) with strong precision, recall, and F1-score values, indicating reliable performance in identifying clause-level perspectives. Furthermore, the summarization component produced coherent and informative outputs, as reflected by ROUGE scores (ROUGE-1: 0.4204, ROUGE-2: 0.2367, ROUGE-L: 0.3303). These results confirm that the system successfully captures essential legal information while maintaining clarity and structure.

In addition to quantitative performance, qualitative evaluation through usability testing highlights the system's ability to significantly improve user comprehension. The integration of plain-language transformation and visualization techniques enables non-expert users to interpret legal documents more easily, supporting better decision-making and reducing dependency on legal expertise.

Despite its effectiveness, the system has certain limitations. The use of predefined perspective categories may not cover all possible legal contexts, and the aggregation-based summarization approach may lack flexibility compared to fully generative models. Nevertheless, the framework provides a practical and reliable solution for enhancing legal document understanding.

Overall, this research demonstrates that combining classification, structured summarization, and visualization offers a powerful approach to addressing the challenges of legal text complexity, contributing toward more accessible and user-centered legal AI systems.

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APPENDICES

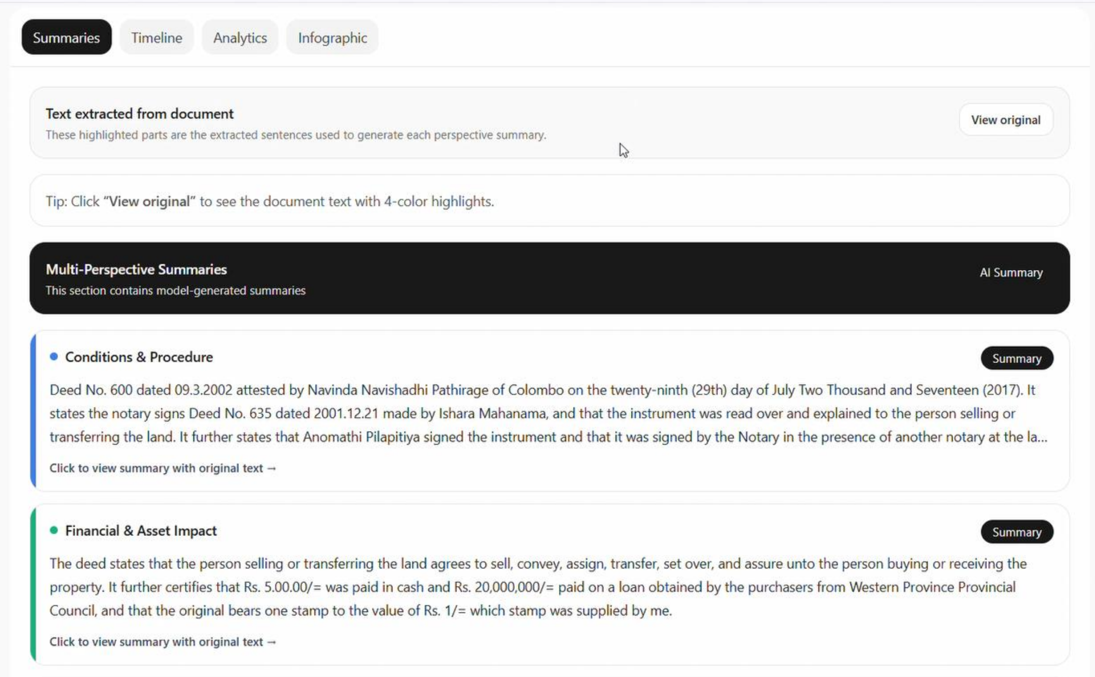
Appendix A: Sample Input Legal Document (Excerpt)

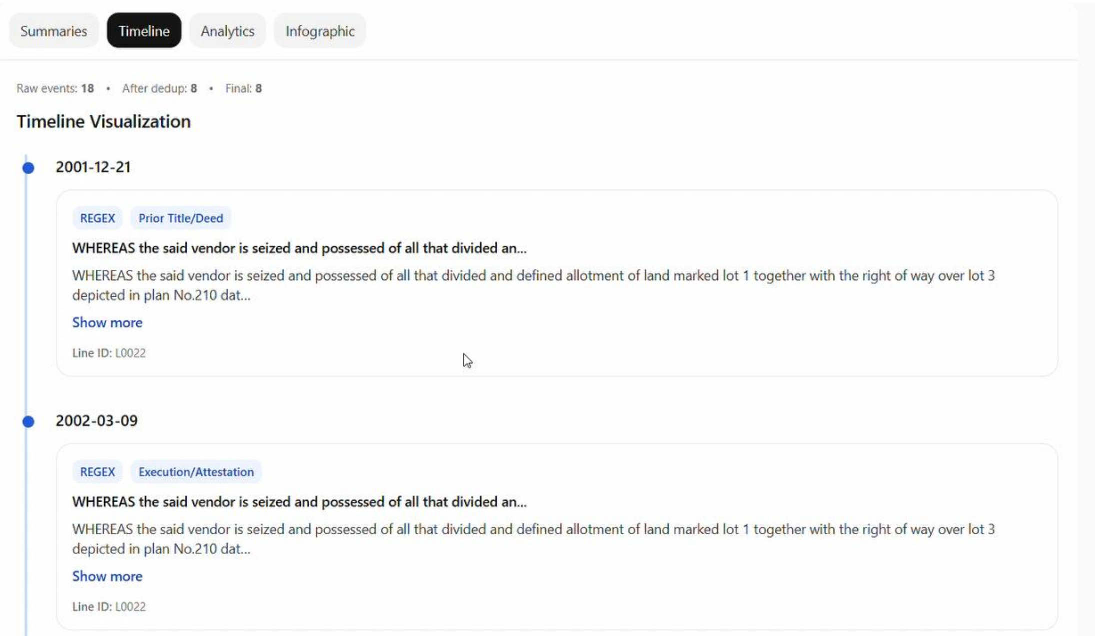
Sample Deeds

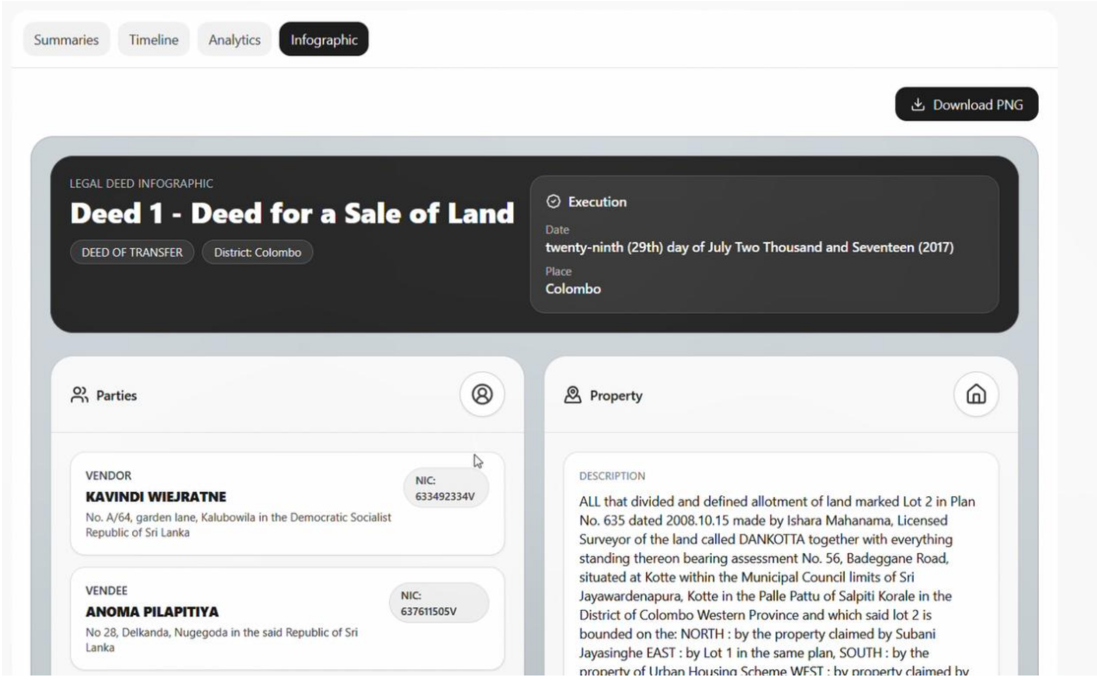
Appendix B: Legal Jargon Dictionary (Sample)

Legal Jargon Dictionary

Appendix C: Screenshots of System Output / UI






Summaries
Timeline
Analytics
Infographic
Download PNG
LEGAL DEED INFOGRAPHIC
Deed 1 - Deed for a Sale of Land
DEED OF TRANSFER
District: Colombo
Execution
Date
twenty-ninth (29th) day of July Two Thousand and Seventeen (2017)
Place
Colombo
Parties
VENDOR
KAVINDI WIEJRATNE
No. A/64, garden lane, Kalubowila in the Democratic Socialist Republic of Sri Lanka
NIC: 633492334V
VENDEE
ANOMA PILAPITIYA
No 28, Delkanda, Nugegoda in the said Republic of Sri Lanka
NIC: 637611505V
Property
DESCRIPTION
ALL that divided and defined allotment of land marked Lot 2 in Plan No. 635 dated 2008.10.15 made by Ishara Mahanama, Licensed Surveyor of the land called DANKOTTA together with everything standing thereon bearing assessment No. 56, Badeggane Road, situated at Kotte within the Municipal Council limits of Sri Jayawardenapura, Kotte in the Palle Pattu of Salpiti Korale in the District of Colombo Western Province and which said lot 2 is bounded on the: NORTH : by the property claimed by Subani Jayasinghe EAST : by Lot 1 in the same plan, SOUTH : by the